Open-Source Car Head Unit Project Outline

Background:

In the world of aftermarket car head units, there are many shortcuts taken by manufacturers in order to ship a well-rounded and reliable product. The market that these aftermarket head units target is significantly composed of people who have a similar interest in modding and customizability in their audio setups.

Typically, people within the car audio community look to these aftermarket head units to expand the capabilities of their car audio systems when OEM systems do not offer the capabilities they want. We believe there is a significant portion of this market that is one step beyond the typical aftermarket head unit in terms of customization needs and would like a fully open-source head unit that they could build upon to suit whatever needs they have.

Goal:

Therefore, the high-level goal of this project is to create an open-source head unit that offers the features of a typical aftermarket head unit, along with additional features such as additional lighting and signal processing capabilities. The structure of these features should be highly modular, allowing for user- and community-created plugins to be added to the head unit.

Specific Design Requirements/Constraints:

The head unit must:

- Run standalone
- Be capable of installation in a standard double-DIN head unit slot
- Support multiple audio inputs, including but not limited to:
  - 1/8" analog input
  - Wi-Fi protocol
  - Bluetooth
  - USB
- Include a touchscreen interface that fits within the double-DIN housing (may or may not include physical buttons)
- Draw no more than 20 watts (not including external modules such as lighting or auxiliary amplifiers)
- Output at least 6 audio channels
- Support DSP of audio channel output, with a minimum of EQ capability
- Offer audio visualization in some capacity
- Be reproducible by anyone with technical expertise and access to the repository

- Integrate required CAN functionality

Additional Non-Required Capabilities:

Some additional capabilities the project can, but is not required to, include are:

- Additional hardware configurations (e.g., options for primary/auxiliary hardware)
- VST support
- Audio visualization through ambient lighting
- Plugin platform/framework
- Video display for movies
- CarPlay integration (if allowed by Apple's licensing)
- Any additional smartphone integration (texts, calls, maps, etc.)
- Custom PCB

Anticipated Technical Approaches and Tools:

- Qt Framework
  - C++
  - QML
- DynamoDB
- Ansible for IaC
- Linux configuration on an SBC
- Potentially JUCE framework for VST integration
- Potentially PCB design

Project Deliverables:

The central deliverable is a fully functional open-source car head unit with all of the minimum requirements listed above. This head unit should be able to be installed in a vehicle, but a demo environment should additionally be created that allows for all head unit capabilities to be demonstrated.

A secondary aspect of this project is its open-source nature. As stated in the requirements, it should be fully reproducible by someone with technical expertise. Thorough documentation should therefore be delivered within the repository covering not only how to build the project, but also the software and hardware design philosophies, constraints, and structure, so that anyone can gain a solid understanding of the project for their own modding needs. The format of this documentation will be left open-ended as long as it meets these requirements.

The open-source nature also requires the team to provide both the source code for the primary application and configuration of the hardware through some form of IaC, as well as plans for the hardware, including hardware lists and wiring diagrams.